

**A GPS-BASED MATERNAL EMERGENCY RESPONSE AND MONITORING
SYSTEM FOR IMPROVING HEALTH IN UGANDA**

CASE STUDY: MUKONO DISTRICT

BY

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DECLARATION

We, **Muhinda Praise Francis, Nabukeera Zaharah Ramulah, and Kato Hassan**, declare that the work presented in this research project proposal is our original work and has not been submitted to any University or Institution of Higher Learning for any academic award. All work from other authors has been fully and properly acknowledged and cited.

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APPROVAL

This is to certify that this research project/report titled: “*A GPS-Based Maternal Emergency Response and Monitoring System for Improving Health in Uganda*” has been carried out under my/our supervision and is now ready for submission to the Examinations Board and Senate of Kyambogo University.

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LIST OF ACRONYMS

GIS: Geographic Information System

WHO: World Health Organization

UBOS: Uganda Bureau of Statistics

IoT: Internet of Things

PPH: Postpartum Haemorrhage

NDP III: Third National Development Plan

SDG: Sustainable Development Goals

GSM: Global System for Mobile Communications

USSD: Unstructured Supplementary Service Data

mHealth: Mobile Health

HIS: Health Information Systems

SMS: Short Message Service

VHTs: Village Health Teams

UNICEF: United Nations Children's Fund

GPS: Global Positioning System

(CEmONC): Comprehensive Emergency Obstetric and Newborn Care

DEFINITION OF TERMS USED

- **GPS (Global Positioning System):** A satellite-based navigation system used to provide precise, real-time geographic coordinates (latitude and longitude) to identify a mother's exact physical location during an emergency.
- **Maternal Health:** The health of women during pregnancy, childbirth, and the postpartum period. In your study, this specifically focuses on reducing mortality and addressing life-threatening complications like Postpartum Haemorrhage.
- **Emergency Response:** The coordinated process of dispatching medical assistance such as ambulances and Village Health Teams (VHTs) to a patient in immediate need of life-saving care.
- **Monitoring System:** A digital infrastructure consisting of a mobile interface and a centralized hospital dashboard used to track, observe, and manage the status and location of expectant mothers in real-time.
- **Spatial Invisibility:** A technical gap where healthcare providers lack digital visibility of a patient's location, often due to poorly mapped areas or reliance on vague verbal directions.
- **Second Delay:** The critical logistical time lost between the decision to seek care and the actual arrival of the mother at a health facility.
- **Maternal Health Informatics:** The interdisciplinary study of the design, development, adoption, and application of IT-based innovations in healthcare services for expectant mothers.
- **Real-time Analytics:** The use of data and resources as soon as they enter a system, allowing for immediate feedback and decision-making during an obstetric crisis.
- **Internet of Things (IoT):** A network of physical objects such as wearable sensors embedded with technologies for the purpose of connecting and exchanging data with other devices over the internet.
- **Geospatial Integration:** The process of combining GPS coordinates with existing health databases to visualize patient locations on a digital map.
- **Referral Loop:** A systemic failure where a patient is sent between multiple health facilities that all lack the necessary resources to treat them, caused by a lack of real-time resource visibility.

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CHAPTER ONE: INTRODUCTION

1.0 Introduction

This chapter provides the foundational framework for the proposed research, which focuses on developing a GPS-integrated maternal health system to reduce mortality rates in Mukono district, Uganda. It begins by establishing the background of the study, contextualizing the maternal health crisis and the critical role of spatial technology in modern healthcare. It then transitions to the problem statement, which critically analyzes the causes and effects of spatial invisibility within current emergency maternal care workflows. Following the problem statement, the study's primary and specific objectives are defined to outline the technical milestones of the research. Furthermore, the scope is established to provide clear subject, geographical, and time boundaries for the project and finally the significance of the study is then presented through a detailed justification of its practical and theoretical contributions to the field of health informatics.

1.1 Background of the study

The field of maternal health informatics has transitioned from static record-keeping to a dynamic integration of the Internet of Things (IoT), Global Positioning Systems (GPS), and real-time analytics. Globally, the World Health Organization (WHO) reports approximately 800 daily maternal deaths from preventable causes, with 95% occurring in low-resource settings (World Health Organization, 2023). In emerging economies like India, the deployment of GPS-tracked ambulances and IoT-enabled wearable sensors has proven effective in reducing the Second Delay, the critical interval required to reach a health facility (Thaddeus & Maine, 1994).

In Africa, digital health initiatives have primarily centred on mHealth. While programs like Kenya's Linda Mama have increased service uptake, they remain fundamentally location-blind. In Uganda, where the maternal mortality ratio stands at 284 deaths per 100,000 live births (**Uganda Bureau of Statistics, 2022**), existing interventions such as UNICEF's SMS-based reminders and Kampala's eHealth referral pilots lack real-time geospatial integration. This technical deficit prevents healthcare providers from tracking patients in labour or identifying precise coordinates during obstetric emergencies.

The current influence of GPS and IoT on Ugandan maternal health is restricted to high-level urban pilots, leaving a significant technological gap in peri-urban and rural districts. Existing systems function as static databases rather than active emergency dispatch tools, resulting in fatal navigation errors during complications like Postpartum Haemorrhage (PPH).

Mukono District, the case study for this research, epitomizes these challenges. As a region characterized by poorly mapped rural villages in Nama and isolated island communities in Koome, verbal directions are frequently unreliable. Consequently, maternal referrals to Mukono General Hospital suffer from spatial invisibility. This study aims to develop a GPS-enabled solution within Mukono to provide a scalable, data-driven model for reducing maternal mortality through precise, real-time location tracking.

1.2 Problem Statement

Despite national interventions, Uganda's maternal mortality ratio remains critically high at 284 deaths per 100,000 live births (**Uganda Bureau of Statistics, 2022**). In Mukono District, a primary driver of these fatalities is the Second Delay, the time lost in locating and transporting a mother during an obstetric emergency. Current response workflows are location-blind, relying on manual records and vague verbal directions, which prevents the timely dispatch of ambulances and Village Health Teams (VHTs). This lack of spatial visibility frequently turns manageable complications, such as Postpartum Haemorrhage, into avoidable tragedies.

The root cause of this crisis is spatial information invisibility. The absence of a digital infrastructure to transmit precise GPS coordinates leads to fatal navigation errors in poorly mapped areas like Nama and isolated island communities like Koome. This is exacerbated by fragmented communication and the reliance on physical paper cards, which restricts access to vital clinical history during transit. Consequently, mothers lose the critical therapeutic window for intervention, often falling into inefficient referral loops between facilities that lack necessary resources. This systemic failure results in unassisted births and high risks of sepsis, primarily affecting rural mothers in Mukono and hindering the achievement of Sustainable Development Goal 3 (**United Nations, 2015**).

To address these challenges, this research proposes a GPS-Enabled Maternal Health Information System for Mukono District. By integrating a mobile interface for mothers with a web-based hospital dashboard, the system will utilize GIS technology to provide real-time tracking and automated resource identification. This solution ensures a mother's precise location is visible to medical responders before she arrives at the facility, effectively bridging the spatial gap in emergency maternal care.

1.3 Objectives

1.3.1 General Objective

The main objective of this research is to design and develop a GPS-Enabled Maternal Health Information System for Mukono District to eliminate spatial invisibility and reduce the Second Delay during obstetric emergencies.

1.3.2 Specific Objectives

- (i) To study and analyse the strengths and weaknesses of the current maternal emergency response and referral workflows within Mukono General Hospital and its surrounding health centres.
- (ii) To determine and specify the functional and technical requirements for a real-time geospatial tracking and alert system tailored for the terrain of Mukono District.
- (iii) To design a GPS-integrated maternal health system consisting of a mobile emergency interface for mothers and a centralized navigation dashboard for hospital administrators.
- (iv) To implement, test, and validate the new system through simulated emergency scenarios to measure its effectiveness in improving response times and coordination.

1.4 Research Questions

To achieve the objectives of this research, the following questions will be addressed;

What are the current strengths and weaknesses of the existing maternal emergency response and referral workflows in Mukono General Hospital and its surrounding health centers?

What are the specific functional and technical requirements necessary to develop a real time geospatial tracking alert system that accounts for the unique terrain of Mukono District?

How can GPS-integrated system be designed to provide an intuitive mobile interface for expectant mothers and a comprehensive navigation?

How efficient is the implemented system in reducing emergency response times and improving coordination when tested through simulated maternal emergency scenarios?

1.5 Scope

1.5.1 Subject Scope

The subject scope of this research focuses on the design and development of a GPS-Enabled Maternal Health Information System. The system is to be developed to address the spatial invisibility of expectant mothers during obstetric emergencies, specifically targeting the reduction of the Second Delay that is time taken to reach a health facility. The primary stakeholders of the proposed system include:

- **Expectant Mothers:** The end-users who require emergency tracking and rapid response.
- **Healthcare Providers:** Midwives and administrators at Mukono General Hospital who will monitor the real-time navigation dashboard.
- **Emergency Responders:** Ambulance drivers and Village Health Teams (VHTs) who will utilize the GPS data for precise navigation.
- **District Health Officers:** Who will use the generated geospatial data for better resource allocation and health planning.

1.5.2 Geographical Scope

The research will be conducted within Mukono District, Central Uganda. This case study was chosen due to its unique geographical diversity, featuring rapidly urbanizing centres alongside poorly mapped rural sub-counties like Nama and isolated island communities in Koome on Lake Victoria. Mukono exemplifies the referral gap where high maternal populations face significant transit hurdles to reach Mukono General Hospital. The district's varied terrain provides a rigorous environment to test the accuracy and reliability of GPS and GIS technologies in low-resource, high-stakes settings.

1.5.3 Time Scope

This research is scheduled to be conducted over a period of 10 weeks, structured into three intensive technical phases:

Weeks 1–3: Research and Requirements. This phase involves the literature review, gathering functional requirements from stakeholders in Mukono, and finalizing the system architecture.

Weeks 4–7: System Development. This phase focuses on the core technical implementation, including geospatial database design, mobile interface coding, and the integration of GPS and Mapping modules.

Weeks 8–10: Validation and Documentation. The final phase is dedicated to system testing, user validation within Mukono District, and the completion of the final project submission.

1.5 Justification

The urgent necessity of this study is underscored by the reality that maternal mortality in Uganda is not merely a medical failure, but a logistical and information failure. While medical protocols for treating complications exist, the inability to locate and reach a mother in distress remains a critical bottleneck. This research is essential because it moves maternal health from a reactive paper-based model to a proactive spatial-awareness model.

Current maternal health interventions in Uganda primarily rely on mHealth (Mobile Health) platforms that focus on one-way communication, such as SMS reminders for antenatal visits. However, a significant technical gap exists, these systems are blind to the mother's physical location during a crisis.

Most existing studies focus on clinical causes of death but overlook the Geospatial Informatics required to solve the Second Delay which is reaching the facility (Faujdar & Kumar, 2020). There is currently no integrated platform in Uganda's public hospitals that links a mother's real-time GPS coordinates to a centralized hospital dashboard for immediate emergency dispatch.

The implementation of this system offers immediate, tangible improvements to the maternal healthcare process:

- **Precision Emergency Response:** By replacing vague verbal directions with precise GPS coordinates, the system minimizes the time ambulances or Village Health Teams (VHTs) spend searching for a patient's home.
- **Intelligent Referral Mapping:** The system can identify the nearest facility with available beds and the correct blood type, preventing the common referral merry-go-round where mothers are bounced between full hospitals.
- **Data-Driven Policy:** Aggregated GPS data will allow the Ministry of Health to identify mortality hotspots, enabling them to place new clinics or ambulances exactly where they are needed most.

This study contributes to the academic body of knowledge in the following ways:

- **GIS-Healthcare Integration:** It provides a new framework for applying Geographic Information Systems to maternal health in resource-constrained environments.
- **Low-Bandwidth Optimization:** Academically, this research explores the synchronization of spatial data over GSM/USSD networks, providing a model for how complex GPS data can be transmitted in areas with poor internet connectivity.
- **Informatics Evolution:** It advances the study of Health Information Systems (HIS) by shifting the focus from static data storage to real-time, event-driven architecture.

This research serves as a pioneer for Spatial Health Informatics in East Africa. By successfully integrating GPS tracking into public hospital workflows, this project provides a scalable template that can be adapted for other emergencies, such as accident response or epidemic tracking. It directly supports the Uganda National Development Plan (NDP III) (**National Planning Authority, 2020**), proving that homegrown, location-aware technology is the most effective tool for reducing the national maternal mortality ratio.

1.6 Chapter Summary

Chapter One has established the foundational framework for this research proposal by identifying the urgent need for a spatial intervention in Uganda's maternal healthcare. The Problem Statement highlighted the fatal consequences of delayed emergency response and the inefficiency of current paper-based or non-localized systems. The Objectives of the study were clearly defined, focusing on the design, development, and testing of a GPS-tracking module and a centralized maternal database.

Furthermore, the Scope and Justification emphasized the technical feasibility and the potential socio-economic impact of the system in alignment with the Uganda National Development Plan (NDP III) and Sustainable Development Goal 3. Ultimately, this chapter sets the stage for the subsequent review of existing literature and the detailed methodology that will be employed to bring this life-saving technical solution to fruition.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents a critical review of literature related to the design and development of a GPS-based maternal emergency response and monitoring system, with specific focus on Mukono District, Uganda. The review examines theoretical and empirical studies in maternal health informatics, situates maternal mortality within both global and national contexts, and analyses the role of geospatial technologies in emergency obstetric care.

Particular emphasis is placed on the interaction between geography, infrastructure, and health system responsiveness in Mukono District. The district's rural sub-counties, island communities, and referral pathways to Mukono General Hospital are examined as key determinants of emergency response efficiency. The chapter concludes by synthesizing the literature to identify gaps that justify the development of a real-time GPS-enabled maternal emergency system.

2.2 Conceptual Framework: Maternal Health Informatics and the Three Delays Model

2.2.1 Evolution of Health Information Systems in Maternal Care

Health Information Systems (HIS) have evolved from static administrative tools into dynamic platforms that support real-time decision-making. Early systems primarily focused on patient registration and aggregate reporting, limiting their utility in emergency scenarios. In Uganda, the Health Management Information System (HMIS) reflects this limitation, as it largely supports retrospective data analysis rather than real-time intervention.

The emergence of mobile health (mHealth) technologies has significantly improved communication in maternal healthcare. Initiatives such as UNICEF's mTrac and MamaUgangi have enhanced antenatal care attendance and drug monitoring through SMS-based platforms. However, these systems remain fundamentally location-blind, lacking the ability to provide spatial intelligence during emergencies.

Recent studies highlight that while mHealth improves information dissemination, it does not address geospatial coordination, which is critical in emergency response. In Mukono District, Village Health Teams (VHTs) rely heavily on mobile phones, yet the absence of location-aware technologies limits their effectiveness in guiding emergency responders.

This indicates a clear technological gap: the transition from communication-based systems to location-aware, real-time emergency systems remains largely unaddressed.

2.2.2 The Three Delays Model

The Three Delays Model provides a foundational framework for understanding maternal mortality. It identifies three critical barriers:

1. Delay in deciding to seek care
2. Delay in reaching a health facility
3. Delay in receiving adequate care

While all three delays are significant, this study focuses on the second delay, which is heavily influenced by geographical and infrastructural factors.

In the context of Mukono District, the second delay is compounded by what this study conceptualizes as spatial invisibility the inability of healthcare providers to accurately locate patients during emergencies. This transforms distance from a physical barrier into an information failure.

Empirical evidence from Uganda shows that delays in reaching facilities are a major contributor to maternal mortality, particularly in rural and peri-urban settings. In Mukono, poorly mapped settlements such as Nama and isolated island communities like Koome intensify this challenge.

Thus, addressing the second delay requires not only transport solutions but also geospatial intelligence systems.

2.3 Maternal Health Context in Mukono District

2.3.1 Geographical and Demographic Characteristics

Mukono District is located in Central Uganda and exhibits significant geographical diversity, including urban centres, rural settlements, and island communities on Lake Victoria. This diversity creates uneven access to healthcare services.

With an estimated population of approximately 700,000, the district has a high proportion of women of reproductive age, increasing demand for maternal health services.

However, geographical constraints particularly in island communities introduce systemic delays in accessing care. Transport limitations, long distances, and unreliable communication networks collectively increase the risk of maternal complications and mortality.

2.3.2 Health Facilities and Referral System

Mukono General Hospital serves as the primary referral facility for comprehensive emergency obstetric and newborn care (CEmONC). Lower-level facilities provide basic services but often lack critical resources such as surgical capacity and blood transfusion services.

The referral system is largely manual and fragmented, relying on phone communication and informal directions. This results in inefficiencies, particularly during emergencies.

Studies indicate that ambulance drivers may spend up to 60 minutes locating patients due to unclear directions. This delay highlights a key weakness: the absence of structured, location-based referral systems.

2.3.3 Village Health Teams (VHTs) and Their Role

Village Health Teams play a crucial role in bridging communities and formal healthcare systems. They facilitate referrals, monitor pregnancies, and provide health education.

Despite their importance, VHTs face significant operational challenges, including lack of digital tools and limited logistical support. Their reliance on verbal communication further contributes to spatial inefficiencies.

Integrating VHTs into a GPS-enabled system would significantly enhance their ability to coordinate emergency responses and improve maternal outcomes.

2.4 Review of GPS and GIS Applications in Maternal Health

2.4.1 Global Evidence on Geospatial Technologies in Emergency Care

Geospatial technologies such as GIS and GPS have transformed emergency response systems globally. They enable real-time tracking, route optimization, and spatial analysis of healthcare access.

In high-income countries, GPS-enabled ambulance systems have significantly reduced response times. Additionally, GIS has been used to identify underserved populations and optimize healthcare resource allocation.

These applications demonstrate that location intelligence is a critical component of effective healthcare delivery.

2.4.2 GPS-Enabled Maternal Health Initiatives in Low-Resource Settings

In sub-Saharan Africa, several pilot projects have explored GPS-based maternal health solutions.

In Tanzania, GPS integration reduced response times by approximately 18 minutes. In Nigeria, the MNETI initiative achieved a significant reduction in transport time. In Kenya, GPS systems improved referral coordination despite infrastructural challenges.

These studies demonstrate the feasibility and effectiveness of GPS technologies, even in low-resource environments. However, most projects remain pilot-based and lack long-term scalability.

2.4.3 Challenges of GPS Integration in Rural Uganda

Despite its potential, GPS implementation in Mukono faces several constraints:

- Limited network coverage, especially in island areas
- Low smartphone penetration
- Data privacy and security concerns
- Dependence on donor-funded pilot projects

These challenges highlight the need for a context-aware system design, incorporating offline functionality, low-bandwidth optimization (e.g., USSD/SMS), and sustainable deployment models.

2.5 Existing Maternal Health Information Systems in Uganda and Mukono

2.5.1 National Systems

Uganda's HMIS and eLMIS systems support health data management and logistics. However, they are not designed for real-time emergency coordination.

2.5.2 District-Level Practices

At the district level, communication is largely informal, often relying on phone calls and messaging platforms such as WhatsApp. These methods lack structure, data integration, and spatial capabilities.

Currently, no system in Mukono District supports:

- Real-time patient tracking
- Automated ambulance dispatch
- Geospatial decision-making

2.5.3 Role of NGOs and Private Sector

NGOs support maternal health through transport schemes and awareness programs. However, these initiatives are fragmented and not integrated into a unified system.

A centralized GPS-based platform could enhance coordination and improve efficiency across stakeholders.

2.6 Spatial Invisibility in Mukono: A Critical Gap

Spatial invisibility emerges as a central challenge in maternal healthcare delivery in Mukono District. It refers to the inability of healthcare providers to accurately identify and navigate to a patient's location.

This issue is driven by:

- Lack of formal addressing systems
- Poorly mapped rural areas
- Geographical isolation of island communities
- Absence of real-time location tracking systems

Studies show that significant time is lost locating patients before transport begins. In emergency conditions such as postpartum hemorrhage, these delays can be fatal.

Thus, spatial invisibility represents a critical intersection between geography, technology, and healthcare failure.

2.7 Synthesis and Research Gap

The reviewed literature reveals several key insights:

- The second delay is a major contributor to maternal mortality in Uganda
- Existing mHealth systems lack geospatial capabilities
- GPS technologies are effective but underutilized in low-resource settings
- Mukono District presents unique geographical challenges

Despite these findings, there is no integrated, GPS-based maternal emergency response system tailored to Mukono District.

This study addresses this gap by proposing a system that:

- Integrates mobile and web platforms
- Enables real-time location tracking
- Supports coordinated emergency response
- Is optimized for low-resource environments

2.8 Chapter Summary

This chapter has critically reviewed literature on maternal health informatics, the Three Delays Model, and geospatial technologies in healthcare. It has highlighted the limitations of existing systems and the unique challenges faced in Mukono District.

The concept of spatial invisibility has been identified as a key barrier to effective emergency response. The absence of real-time, location-aware systems represents a significant research and practical gap.